# Multi-RIS-Aided User Localization With Virtual Line-of-Sight Paths

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***Abstract*—Reconfigurable intelligent surface (RIS) is an emerging technology for future wireless communication networks. The virtual line-of-sight (LoS) path introduced by RIS not only enhances wireless communication capability but also can be used to improve positioning accuracy in non-LoS environments. In this paper, we consider user localization in multi-RIS-aided systems with virtual LoS paths, but without any prior information of the user location. To recover the channel parameters of virtual LoS paths for user localization, we formulate the received signals at base station as a three-dimensional tensor in a CANDECOMP/PARAFAC (CP) decomposition form corrupted by Gaussian noise. Then the alternating least squares (ALS) is adopted to infer the factor matrices in the CP decomposition. With the converged factor matrices, the multiple signal classification (MUSIC) algorithm is adopted to extract the angle and path delay information of the virtual LoS paths. Using the angle and path delay information, a closed-form estimation of the user location is derived. Numerical results demonstrate that the proposed method achieves superior performance in estimating virtual LoS channel parameters and the user location.**

***Index Terms*—Reconfigurable intelligent surface, virtual LoS paths, user localization, alternating least squares.**

## I. Introduction

Reconfigurable intelligence surface (RIS) [1]–[3] is becoming a key technology to improve wireless communication performance in 6G. RIS consists of passive arrays of programmable elements with low energy consumption, which can be controlled to reflect or refract electromagnetic waves along desired directions. Even when the direct LoS path between transmitter and receiver is blocked, RIS can provide a virtual LoS path for wireless communications, enhancing the coverage, energy efficiency and spectral efficiency of wireless communication systems. On the other hand, a RIS with known location can be treated as an anchor for localization in RIS-aided systems. Compared to the traditional localization systems, extra measurements from the virtual LoS path, such as time of arrival (ToA), angle of arrival (AoA) or departure (AoD), and received signal strength (RSS), can be used to determine the user location in non-LoS conditions [4]–[6].

The promise of RIS in localization has been demonstrated in a few research works in single-RIS-aided systems [7]–[11]. However, the coverage of single RIS could be quite limited in complex environment, where both direct LoS path and the virtual LoS path may not exist. To this end, it is desirable to deploy multiple RISs simultaneously in one system. Currently, only several research works have considered the localization in multi-RIS-aided scenarios [12]–[15]. In [12], a joint beam training and positioning method in multi-RIS-aided millimeter wave communications systems is proposed, where the LoS path and the virtual LoS paths are decoupled by first deactivating all RISs and then activating only one RIS at each timeslot. This method requires the extra control of all RISs one by one, which is not time-efficient. In [13], an approximate maximum likelihood (AML) method is proposed in the scenario with multiple RISs reflecting the signal simultaneously. The channel parameters of LoS path and the virtual LoS paths are estimated first, and then an energy-based method is adopted to match the channel parameters with each path. In order to leverage the prior distribution of the user location, a message passing algorithm is developed to estimate and track a single user's location in multi-RIS aided signal-carrier MIMO systems [14]. To further extend to multi-RIS-aided multiuser scenarios, a Bayesian multiuser tracking algorithm is proposed to estimate the users' locations at different frames in [15].

Against this background, we consider user localization in multi-RIS-aided systems with virtual LoS paths, but without any prior information of the user location. First, we formulate the received virtual LoS path signal at the base station (BS) as a classical CANDECOMP/PARAFAC (CP) tensor decomposition form corrupted by Gaussian noise. Then, the alternating least squares (ALS) is adopted to estimate the factor matrices. To further estimate the virtual LoS channel parameters from the converged factor matrices, the multiple signal classification (MUSIC) algorithm is applied to extract the angle and path delay information. Using the angle and path delay information associated to multiple RISs, the user location is estimated in a closed form. Finally, numerical results are presented to demonstrate the performance of the proposed method in estimating the virtual LoS channel parameters and the user location, with comparisons against the AML method.

## II. System Model and Problem Formulation

Consider a multi-RIS-aided system consisting of a multi-antenna BS, a single-antenna user, and $M$ RISs. The BS is

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deployed with $N_B$ antennas, and each RIS consists of $N_R$ reflective elements. The antennas of the BS and reflective elements of each RIS are configured in uniform linear arrays. The spacing between antennas and that between reflective elements are set to one half of the carrier wavelength. In this paper, we consider the uplink transmission of a multi-RIS-aided orthogonal frequency division multiplexing (OFDM) system for user localization. It is assumed that the line-of-sight (LoS) path between the BS and the user is blocked. In order to realize user localization, the BS receives an OFDM pilot signal transmitted by the user with the aid of multiple RISs.

More specifically, the uplink channel $\mathbf{h}_{k,m}$ from the user to the $m$-th RIS and $\mathbf{G}_{k,m}$ from the $m$-th RIS to the BS over the $k$-th OFDM subcarrier are modelled as

$$\mathbf{h}_{k,m} = \mathbf{h}_{k,m}^{LoS} + \mathbf{h}_{k,m}^{NLoS}, \tag{1}$$

$$\mathbf{G}_{k,m} = \mathbf{G}_{k,m}^{LoS} + \mathbf{G}_{k,m}^{NLoS}, \tag{2}$$

where $\mathbf{h}_{k,m}^{LoS}, \mathbf{h}_{k,m}^{NLoS}$ denote the LoS component and non-LoS component of the channel from the user to the $m$-th RIS over the $k$-th subcarrier, respectively. Similarly, $\mathbf{G}_{k,m}^{LoS}, \mathbf{G}_{k,m}^{NLoS}$ denote the LoS component and non-LoS component of the channel from the $m$-th RIS to the BS over the $k$-th subcarrier, respectively. For the LoS components $\mathbf{h}_{k,m}^{LoS}$ and $\mathbf{G}_{k,m}^{LoS}$, they can be modelled as

$$\mathbf{h}_{k,m}^{LoS} = \rho_{U,m} e^{-j\frac{2\pi f_s(k-1)}{K}\tau_{U,m}} \boldsymbol{\phi}_{N_R}(\sin\theta_{U,m}), \tag{3}$$

$$\mathbf{G}_{k,m}^{LoS} = \rho_{B,m} e^{-j\frac{2\pi f_s(k-1)}{K}\tau_{B,m}} \boldsymbol{\phi}_{N_B}(\sin\theta_m)\boldsymbol{\phi}_{N_R}^H(\sin\psi_m), \tag{4}$$

where $f_s$ denotes the bandwidth, $\rho_{U,m}$ and $\tau_{U,m}$ represent the path loss and path delay corresponding to the LoS path between the user and the $m$-th RIS, respectively, $\rho_{B,m}$ and $\tau_{B,m}$ represent the path loss and path delay corresponding to the LoS path between the $m$-th RIS and the BS, respectively. Furthermore, $\theta_{U,m}$ is the angle of arrival (AoA) from the user to the $m$-th RIS, $\theta_m$ and $\psi_m$ are the AoA and angle of departure (AoD) from the $m$-th RIS to the BS, and the steering vector $\boldsymbol{\phi}_N(v) = [1, e^{j\pi v}, e^{j2\pi v}, \cdots, e^{j(N-1)\pi v}]^T$.

It is assumed that the OFDM transmission contains $K$ subcarriers and $G$ OFDM symbols are transmitted within the coherence time, during which channel coefficients and user position remain unchanged. Moreover, the passive beamforming configuration of the RIS varies over different OFDM symbols. Thus, the received signal at the BS over the $k$-th subcarrier with respect to the $g$-th OFDM symbol can be written as

$$\mathbf{r}_{g,k} = \sum_{m=1}^{M} \mathbf{G}_{k,m}\boldsymbol{\Omega}_{g,m}\mathbf{h}_{k,m}x_{g,k} + \mathbf{w}_{g,k}, \tag{5}$$

where $x_{g,k} = 1$ is the pilot signal, $\boldsymbol{\Omega}_{g,m} \triangleq \mathrm{diag}\left(\frac{1}{\sqrt{N_R}}[e^{j\omega_{1,g,m}}, \ldots, e^{j\omega_{N_R,g,m}}]^T\right)$ with $\boldsymbol{\omega}_{g,m} \triangleq [\omega_{1,g,m}, \omega_{2,g,m}, \cdots, \omega_{N_R,g,m}]^T$ being the phase shifts of the $m$-th RIS over the $g$-th OFDM symbol, and $\mathbf{w}_{g,k} \triangleq [w_{1,g,k}, \ldots, w_{N_B,g,k}]^T$ represents the white Gaussian noise.

By substituting (1), (2), (3) and (4) into (5), we obtain

$$\mathbf{r}_{g,k} = \sum_{m=1}^{M} \rho_{B,m}\rho_{U,m} e^{-j\frac{2\pi f_s(k-1)}{K}(\tau_{B,m}+\tau_{U,m})} \times \boldsymbol{\phi}_{N_B}(\sin\theta_m)\boldsymbol{\phi}_{N_R}^H(\sin\psi_m)\boldsymbol{\Omega}_{g,m}\boldsymbol{\phi}_{N_R}(\sin\theta_{U,m}) + \mathbf{n}_{g,k}, \tag{6}$$

where

$$\mathbf{n}_{g,k} = \sum_{m=1}^{M} \mathbf{G}_{k,m}^{LoS}\boldsymbol{\Omega}_{g,m}\mathbf{h}_{k,m}^{NLoS} + \sum_{m=1}^{M} \mathbf{G}_{k,m}^{NLoS}\boldsymbol{\Omega}_{g,m}\mathbf{h}_{k,m}^{LoS} + \sum_{m=1}^{M} \mathbf{G}_{k,m}^{NLoS}\boldsymbol{\Omega}_{g,m}\mathbf{h}_{k,m}^{NLoS} + \mathbf{w}_{g,k}. \tag{7}$$

Typcially, $\mathbf{n}_{g,k}$ is unknown and assumed to be an isotropic complex Gaussian distribution, i.e., $\mathbf{n}_{g,k} \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I})$, where $\sigma^2\mathbf{I}$ is the covariance matrix.

After receiving the signals $\mathbf{r}_{g,k}$ for all $g = 1, 2, \cdots, G$ and $k = 1, 2, \cdots, K$ at the BS, our goal is to infer the user location. For the user location, it is related to the channel parameters $\tau_{U,m}, \sin\theta_{U,m}$ in the virtual LoS components. According to the geometric relationship, we have

$$\tau_{U,m} = \frac{\|\mathbf{u} - \mathbf{s}_m\|_2}{c}, \tag{8}$$

$$\sin\theta_{U,m} = \frac{u_y - s_{y,m}}{\|\mathbf{u} - \mathbf{s}_m\|_2}, \tag{9}$$

where $c$ is the speed of light, $\|\cdot\|_2$ denotes the Euclidean norm, $\mathbf{s}_m \triangleq [s_{x,m}, s_{y,m}]^T$ and $\mathbf{u} \triangleq [u_x, u_y]^T$ are the locations of the $m$-th RIS and the user, respectively. In order to estimate the user location, we need to recover the channel parameters $\{\tau_{U,m}, \sin\theta_{U,m}\}_{m=1}^M$ corresponding to the virtual LoS paths.

## III. User Localization Using the Virtual LoS Path Parameters

The received signals $\mathbf{r}_{g,k}$ for all $g = 1, 2, \cdots, G$ and $k = 1, 2, \cdots, K$ can be denoted by a third-order tensor $\boldsymbol{\mathcal{R}} \in \mathbb{C}^{N_B \times G \times K}$ with the $(i, g, k)$-th element

$$[\boldsymbol{\mathcal{R}}]_{i,g,k} = \sum_{m=1}^{M} \rho_{B,m}\rho_{U,m} e^{-j\frac{2\pi f_s(k-1)}{K}(\tau_{B,m}+\tau_{U,m})} e^{j\pi(i-1)\sin\theta_m} \times \boldsymbol{\phi}_{N_R}^H(\sin\psi_m)\boldsymbol{\Omega}_{g,m}\boldsymbol{\phi}_{N_R}(\sin\theta_{U,m}) + [\boldsymbol{\mathcal{N}}]_{i,g,k}, \tag{10}$$

where the noise $\boldsymbol{\mathcal{N}} \in \mathbb{C}^{N_B \times G \times K}$ has the $(i, g, k)$-th element $n_{i,g,k} \sim \mathcal{CN}(0, \sigma^2)$. Moreover, we can write the CP decomposition of tensor $\boldsymbol{\mathcal{R}}$ as

$$\boldsymbol{\mathcal{R}} = \sum_{m=1}^{M} \mathbf{a}_m \circ \mathbf{b}_m \circ \mathbf{c}_m + \boldsymbol{\mathcal{N}}, \tag{11}$$

where '$\circ$' denotes the vector outer product,

$$\mathbf{a}_m = [1, e^{j\pi \sin\theta_m}, \cdots, e^{j\pi(N_B-1)\sin\theta_m}]^T, \quad (12)$$

$$\begin{aligned}\mathbf{b}_m = [&\boldsymbol{\phi}_{N_R}^H(\sin\psi_m)\boldsymbol{\Omega}_{1,m}\boldsymbol{\phi}_{N_R}(\sin\theta_{U,m}),\\ &\boldsymbol{\phi}_{N_R}^H(\sin\psi_m)\boldsymbol{\Omega}_{2,m}\boldsymbol{\phi}_{N_R}(\sin\theta_{U,m}),\\ &\vdots\\ &\boldsymbol{\phi}_{N_R}^H(\sin\psi_m)\boldsymbol{\Omega}_{G,m}\boldsymbol{\phi}_{N_R}(\sin\theta_{U,m})]^T,\end{aligned} \quad (13)$$

$$\mathbf{c}_m = \rho_m[1, e^{-j\frac{2\pi f_s}{K}\tau_m}, \cdots, e^{-j\frac{2\pi f_s(K-1)}{K}\tau_m}]^T, \quad (14)$$

with $\tau_m \triangleq \tau_{B,m} + \tau_{U,m}$, and $\rho_m \triangleq \rho_{B,m}\rho_{U,m}$.

In order to recover the factor matrices in the CP decomposition of $\boldsymbol{\mathcal{R}}$, the ALS algorithm [16] can be applied. In ALS, three factor matrices $\mathbf{A} \triangleq [\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_M]$, $\mathbf{B} \triangleq [\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_M]$, $\mathbf{C} \triangleq [\mathbf{c}_1, \mathbf{c}_2, \cdots, \mathbf{c}_M]$ are updated iteratively in a cyclic manner. First, fixing the factor matrices $\mathbf{B}$ and $\mathbf{C}$, $\mathbf{A}$ is updated by

$$\mathbf{A} = \mathbf{R}_{(1)}\left[(\mathbf{C}\otimes\mathbf{B})^T\right]^\dagger, \quad (15)$$

where '$\otimes$' denotes the Khatri–Rao product and '$\dagger$' denotes the Moore–Penrose pseudo-inverse. Then, fixing the factor matrices $\mathbf{A}$ and $\mathbf{C}$, $\mathbf{B}$ is updated by

$$\mathbf{B} = \mathbf{R}_{(2)}\left[(\mathbf{C}\otimes\mathbf{A})^T\right]^\dagger. \quad (16)$$

Finally, fixing the factor matrices $\mathbf{A}$ and $\mathbf{B}$, $\mathbf{C}$ is updated by

$$\mathbf{C} = \mathbf{R}_{(3)}\left[(\mathbf{B}\otimes\mathbf{A})^T\right]^\dagger. \quad (17)$$

In the above equations, $\mathbf{R}_{(n)}$ denotes the mode-$n$ matricization of $\boldsymbol{\mathcal{R}}$. Once the ALS converges or reaches the maximum number of iterations in repeating (15)–(17), the channel parameters of virtual LoS paths can be recovered from the factor matrices.

According to [17], if $k_\mathbf{A} + k_\mathbf{B} + k_\mathbf{C} \geq 2M+2$, the CP decomposition of $\boldsymbol{\mathcal{R}}$ is unique with the exception of permutation indeterminacy and scaling indeterminacy, where $k_\mathbf{A}$ denote the $k$-$rank$ of the matrix $\mathbf{A}$. Thus the estimated factor matrices $\hat{\mathbf{A}}$, $\hat{\mathbf{B}}$, $\hat{\mathbf{C}}$ may have only the permutation indeterminacy and scaling indeterminacy of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, respectively. For the $\ell$-th column of the estimated factor matrices, we have

$$\hat{\mathbf{A}}_{:,\ell} \approx \lambda_{\ell,1}\mathbf{a}_m, \quad (18)$$

$$\hat{\mathbf{B}}_{:,\ell} \approx \lambda_{\ell,2}\mathbf{b}_m, \quad (19)$$

$$\hat{\mathbf{C}}_{:,\ell} \approx \lambda_{\ell,3}\mathbf{c}_m, \quad (20)$$

with the unknown RIS index $m \in \mathcal{M}$ and $\lambda_{\ell,1}\lambda_{\ell,2}\lambda_{\ell,3} = 1$.

Next, we recover the virtual LoS channel parameters for user location estimation. On one hand, since each column of $\mathbf{A}$, $\mathbf{C}$ has the similar form of the steering vector, the multiple signal classification (MUSIC) algorithm [18] is applied to extract $\{\widetilde{\sin\theta_\ell}\}_{\ell=1}^M$ and $\{\widetilde{\tau}_\ell\}_{\ell=1}^M$ based on $\hat{\mathbf{A}}$, $\hat{\mathbf{C}}$, where $\ell$ is the column index of the estimated factor matrices.

Although the virtual LoS channel parameters $\{\widetilde{\sin\theta_\ell}, \widetilde{\tau}_\ell\}_{\ell=1}^M$ have been estimated, these parameter estimations are required to further associate to specific RISs. Since the locations of RISs and BS are known, we can obtain the exact value of $\sin\theta_m$. Based on the true values $\{\sin\theta_m\}_{m=1}^M$ and the estimated values of $\{\widetilde{\sin\theta_\ell}\}_{\ell=1}^M$, we associate the estimated virtual LoS channel parameters to specific RISs. To match the indices of RISs that exist virtual LoS paths from user to BS, we minimize the $\ell_2$ norm error between the true values and estimated values. In other words, we find a mapping $\chi$: $\ell \in \{1,2,\cdots,M\} \to m \in \{1,2,\cdots,M\}$ by minimizing $\sum_{\ell=1}^M \|\widetilde{\sin\theta_\ell} - \sin\theta_{\chi(\ell)}\|_2$, where $\chi(\ell) \in \{1,2,\cdots,M\}$ is the RIS index corresponding to the $\ell$-th column of the estimated factor matrices. Thus we can match the column index of the estimated factor matrices to the RIS index.

On the other hand, the $\ell$-th column of $\mathbf{B}$ can be rewritten as

$$\mathbf{b}_\ell = \boldsymbol{\Pi}_\ell \boldsymbol{\phi}_{N_R}(\widetilde{\sin\theta_{U,\ell}}), \quad (21)$$

where

$$\boldsymbol{\Pi}_\ell = \begin{bmatrix} \boldsymbol{\phi}_{N_R}^H(\sin\psi_{\chi(\ell)})\boldsymbol{\Omega}_{1,\chi(\ell)} \\ \boldsymbol{\phi}_{N_R}^H(\sin\psi_{\chi(\ell)})\boldsymbol{\Omega}_{2,\chi(\ell)} \\ \vdots \\ \boldsymbol{\phi}_{N_R}^H(\sin\psi_{\chi(\ell)})\boldsymbol{\Omega}_{G,\chi(\ell)} \end{bmatrix}. \quad (22)$$

Thus we can first calculate $\boldsymbol{\Pi}_\ell^\dagger \mathbf{M_B}$, and then extract $\widetilde{\sin\theta_{U,\ell}}$ by the MUSIC algorithm.

According to the geometric relationship, we have

$$\widetilde{\sin\theta_{U,\ell}} = \frac{u_y - s_{y,\chi(\ell)}}{\|\mathbf{u} - \mathbf{s}_{\chi(\ell)}\|_2}, \ell = 1,2,\cdots,M, \quad (23)$$

$$\widetilde{\tau}_\ell = \tau_{B,\chi(\ell)} + \frac{\|\mathbf{u} - \mathbf{s}_{\chi(\ell)}\|_2}{c}, \ell = 1,2,\cdots,M, \quad (24)$$

where $\tau_{B,\chi(\ell)} = \|\mathbf{s}_{\chi(\ell)} - \mathbf{b}\|_2/c$ is known. With some further manipulations, we get the closed-form expressions of the user location as

$$u_x = \frac{1}{M-1}\sum_{\ell=2}^M \frac{\Delta d_\ell - 2(s_{y,\chi(1)} - s_{y,\chi(\ell)})u_y}{2(s_{x,\chi(1)} - s_{x,\chi(\ell)})}, \quad (25)$$

$$u_y = \frac{1}{M}\sum_{\ell=1}^M s_{y,\chi(\ell)} + c(\widetilde{\tau}_\ell - \tau_{B,\chi(\ell)})\widetilde{\sin\theta_{U,\ell}}, \quad (26)$$

where $\Delta d_\ell \triangleq c^2(\widetilde{\tau}_\ell - \tau_{B,\chi(\ell)})^2 - c^2(\widetilde{\tau}_1 - \tau_{B,\chi(1)})^2 - s_{x,\chi(\ell)}^2 - s_{y,\chi(\ell)}^2 + s_{x,\chi(1)}^2 + s_{y,\chi(1)}^2$.

## IV. Numerical Results

In the simulations, we consider a multi-RIS-aided MISO-OFDM system with carrier wavelength $\lambda = 10.7\mathrm{mm}$, bandwidth $f_s = 10\mathrm{MHz}$, and the number of OFDM subcarriers $K = 30$. In this system, the total number of RISs is set to $M = 3$, where the locations of RISs are $\mathbf{s}_1 \triangleq (-23,5)\mathrm{m}$, $\mathbf{s}_2 \triangleq (23,13)\mathrm{m}$, $\mathbf{s}_3 \triangleq (33,19)\mathrm{m}$, respectively. Moreover, the location of BS is $\mathbf{b} \triangleq (50,-2)\mathrm{m}$ and the location of user is $\mathbf{u} \triangleq (-50,-6)\mathrm{m}$. The path losses are modelled as $\rho_{U,m} = C_0(\|\mathbf{u} - \mathbf{s}_m\|_2)^{-\iota/2}$ and $\rho_{B,m} = C_0(\|\boldsymbol{b} - \mathbf{s}_m\|_2)^{-\iota/2}$, where $\iota = 2.5$ and $C_0 = -15\mathrm{dB}$. Furthermore, the number of BS antennas $N_B = 5$, the number of RIS reflective elements $N_R = 5$, and the number of OFDM symbols $G = 10$. Here, SNR is defined as $\|\boldsymbol{\mathcal{R}} - \boldsymbol{\mathcal{N}}\|_F^2/\|\boldsymbol{\mathcal{N}}\|_F^2$, where $\|\cdot\|_F$ denotes the Frobenius norm of a tensor.

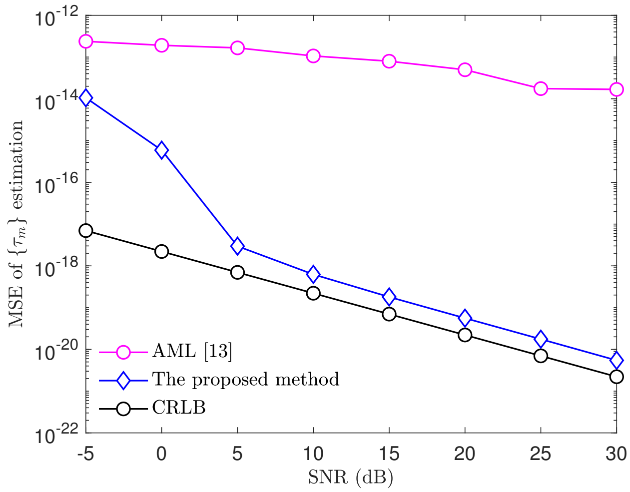


Fig. 1. The MSE of $\{\tau_m\}$ estimation

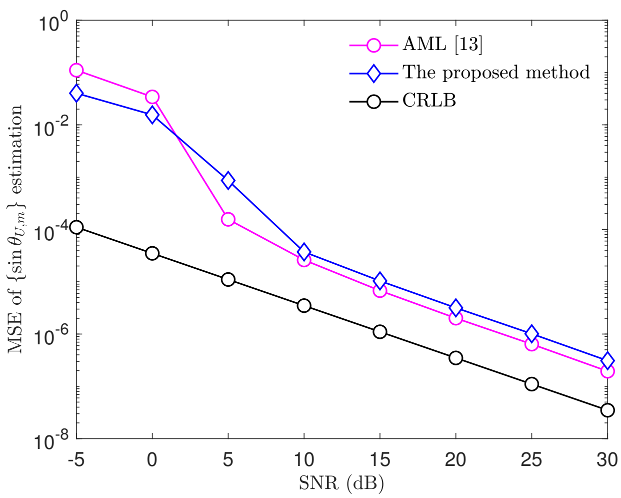


Fig. 2. The MSE of $\{\sin\theta_{U,m}\}$ estimation

It can be verified that $k_{\mathbf{A}} + k_{\mathbf{B}} + k_{\mathbf{C}} \geq 2M + 2$ for $M = 3$, which implies the CP decomposition of $\boldsymbol{\mathcal{R}}$ is unique with the exception of permutation indeterminacy and scaling indeterminacy. Since there is no prior information about the user location, the phase shifts of RISs are randomly initialized and varied over different OFDM symbols. For each OFDM symbol, the phase shifts of RISs are configured by randomly drawn from a uniform distribution over $[0, 2\pi]$, i.e., $\omega_{n,g,m} \sim \mathcal{U}[0, 2\pi]$. In the ALS-based method (the proposed method), the initialization of the factor matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are set to $M = 3$ leading left singular vectors of $\mathbf{R}_{(n)}$ for $n = 1, \cdots, M$, respectively. Furthermore, 2000 Monte Carlo simulations are conducted to evaluate the mean square errors (MSEs) of the channel parameter estimations and the root mean square error (RMSE) of user localization.

Figures 1 and 2 show the MSEs of $\{\tau_m\}$ and $\{\sin\theta_{U,m}\}$ estimations, respectively. The proposed method estimates $\{\tau_m\}$

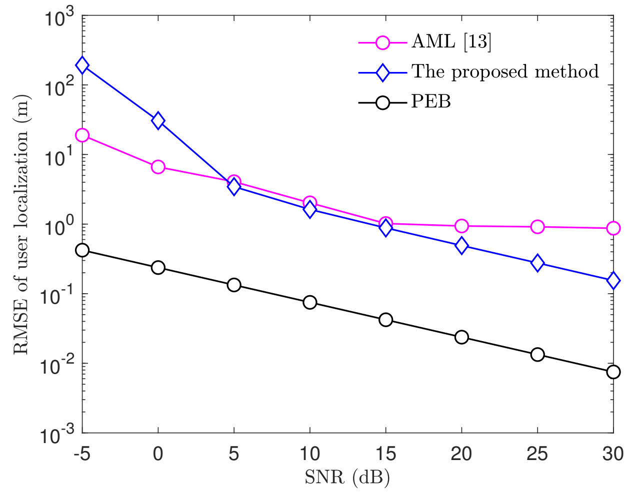


Fig. 3. The RMSE of user localization

much more accurately than the AML method and is closed to Cramér-Rao Lower Bound (CRLB) in high SNRs. This is because the AML method directly adopts the MUSIC algorithm to estimate the channel parameters and the MUSIC algorithm has a poor performance when the sources are coherent. On the other hand, the proposed method achieves smaller error in estimating $\{\sin\theta_{U,m}\}$ than the AML method in low SNRs and close performance to the AML method in high SNRs.

Figure 3 shows the RMSE of location estimation and a comparison to the position error bound (PEB). The PEB demonstrates the localization in this multi-RIS-aided system can achieve centimeter-level accuracy in high SNRs. Although the proposed method cannot reach the PEB with a small number of pilots, it can achieve sub-meter-level localization accuracy in high SNRs. Moreover, it can be observed that the localization accuracy by the proposed method is better than the AML method in high SNRS and closed to the AML method in medium SNRs. Note that if we increase the antennas of BS, the elements of each RIS, or the number of OFDM pilots, these estimation performances will increase.

## V. Conclusions

In this paper, we proposed a user localization method in multi-RIS-aided systems with virtual LoS paths, but without any prior information of the user location. First, we formulated the received signal at the BS in a CP tensor decomposition form corrupted by Gaussian noise. Thus, the ALS algorithm was adopted to estimate the factor matrices in the CP decomposition. With the converged factor matrices, the channel parameters of the virtual LoS paths were extracted by the MUSIC algorithm. Then, the user location was estimated in closed-form using the angle and path delay information of virtual LoS paths associated to multiple RISs. Finally, numerical results were presented to demonstrate the superior performance of the proposed method for virtual LoS channel parameter estimation and user localization.

# REFERENCES


[1] S. Hu, F. Rusek, and O. Edfors, "Beyond massive MIMO: the potential of data transmission with large intelligent surfaces," *IEEE Trans. Signal Process.*, vol. 66, no. 10, pp. 2746–2758, May 2018.

[2] C. Liaskos, S. Nie, A. Tsioliaridou, A. Pitsillides, S. Ioannidis, and I. Akyildiz, "A new wireless communication paradigm through software-controlled metasurfaces," *IEEE Commun. Mag.*, vol. 56, no. 9, pp. 162–169, Sept. 2018.

[3] C. Pan, H. Ren, K. Wang, et al., "Reconfigurable intelligent surfaces for 6G systems: principles, applications, and research directions," *IEEE Commun. Mag.*, vol. 59, no. 6, pp. 14-20, Jun. 2021.

[4] H. Wymeersch, J. He, B. Denis, A. Clemente and M. Juntti, "Radio localization and mapping with reconfigurable intelligent surfaces: challenges, opportunities, and research directions," *IEEE Veh. Technol. Mag.*, vol. 15, no. 4, pp. 52-61, Dec. 2020

[5] T. Ma, Y. Xiao, X. Lei, L. Zhang, Y. Niu and G. K. Karagiannidis, "Reconfigurable intelligent surface-assisted localization: technologies, challenges, and the road ahead," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 1430-1451, Jul. 2023.

[6] A. Umer, I. Müürsepp, M. M. Alam and H. Wymeersch, "Reconfigurable intelligent surfaces in 6G radio localization: a survey of recent developments, opportunities, and challenges," *IEEE Commun. Surv. Tutor.*, Early Access.

[7] J. He, H. Wymeersch, T. Sanguanpuak, O. Silvén, and M. Juntti, "Adaptive beamforming design for mmwave RIS aided joint localization and communication," in *Proc. IEEE Wirel. Commun. Netw. Conf. Workshops*, 2020, pp. 1–6.

[8] H. Zhang, H. Zhang, B. Di, K. Bian, Z. Han, and L. Song, "Towards ubiquitous positioning by leveraging reconfigurable intelligent surface," *IEEE Commun. Lett.*, vol. 25, no. 1, pp. 284–288, Jan. 2021.

[9] Y. Li, Y. Luo, X. Wu, Z. Shi, S. Ma and G. Yang, "Variational Bayesian learning based localization and channel reconstruction in RIS-aided systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 11309-11324, Sept. 2024.

[10] A. Elzanaty, A. Guerra, F. Guidi, and M.-S. Alouini, "Reconfigurable intelligent surfaces for localization: position and orientation error bounds," *IEEE Trans. Signal Process.*, vol. 69, pp. 5386–5402, 2021.

[11] Y. Han, S. Jin, C.-K. Wen, and T. Q. S. Quek, "Localization and channel reconstruction for extra large ris-assisted massive mimo systems," *IEEE J. Sel. Top. Signal Process.*, vol. 16, no. 5, pp. 1011–1025, Aug. 2022.

[12] W. Wang and W. Zhang, "Joint beam training and positioning for intelligent reflecting surfaces assisted millimeter wave communications," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6282–6297, Oct. 2021.

[13] W. Zhang, Z. Wang and W. P. Tay, "Approximate maximum-likelihood RIS-aided positioning," *IEEE Trans. Wireless Commun.*, vol. 22, no. 12, pp. 8859-8875, Dec. 2023.

[14] B. Teng, X. Yuan, R. Wang, and S. Jin, "Bayesian user localization and tracking for reconfigurable intelligent surface aided MIMO systems," *IEEE J. Sel. Top. Signal Process.*, vol. 16, no. 5, pp. 1040–1054, Aug. 2022.

[15] B. Teng, X. Yuan, and R. Wang, "Variational bayesian multiuser tracking for reconfigurable intelligent surface aided MIMO-OFDM systems," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 12, pp. 3752– 3767, Dec. 2023.

[16] N. D. Sidiropoulos, L. De Lathauwer, X. Fu, K. Huang, E. E. Papalexakis, and C. Faloutsos, "Tensor decomposition for signal processing and machine learning," *IEEE Trans. Signal Process.*, vol. 65, no. 13, pp. 3551–3582, Jul. 2017.

[17] J. B. Kruskal, "Rank, decomposition, and uniqueness for 3-way and N-way arrays," in Multiway Data Analysis. Amsterdam, The Netherlands: Elsevier, 1989, pp. 7–18.

[18] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Trans. Antennas Propag.*, vol. 34, no. 3, pp. 276-280, Mar. 1986.